The Korteweg-de Vries Equation

Solitons: An Introduction
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Consider the standard one-dimensional wave equation

$$u_{tt} - c^2 u_{xx} = 0$$

The general solution (see d'Alembert's solution) can be described as

$$u(x,t) = f(x - ct) + g(x + ct),$$

where f and g are arbitrary functions determined by initial conditions. Note that they are not necessarily differentiable. This this is a linear differential equation (i.e. f and g are solutions on their own), we can consider only the waves that travel to the right. In doing so, we restrict the discussion to the solution of

$$u_t + cu_x = 0$$

Which has the general solution

$$u(x,t) = f(x - ct)$$

Note that this new differential equation is a factor in the factorization of the standard one-dimensional linear wave operator. We will consider this "right travelling" wave equation here, for $c = 1$.

The simplest dispersive wave equation (equations governing waves that exhibit frequency dispersion, meaning waves of different wavelengths travel at different phase speeds) is

$$u_t + u_x + u_{xxx} = 0.$$

We show that

$$u(x,t) = e^{i(kx - \omega t)}$$

is a solution to this equation if $\omega = k - k^3$.

$$-i\omega e^{i(kx-\omega t)} + ike^{i(kx-\omega t)} - ik^3 e^{i(kx-\omega t)} = 0$$

$$\implies (\omega - k + k^3)e^{i(kx-\omega t)} = 0$$
$$\implies \omega = k - k^3.$$

$\omega = k - k^3$ is called the dispersion relation - a function $\omega(k)$. We call $k \in \mathbb{R}$ the wave number, and ω the frequency. Since

$$kx - \omega t = k(x - (1 - k^2)t),$$

the solution propagates at a velocity of

$$c = \frac{\omega}{k} = 1 - k^2.$$

We can consider a general solution by integrating over all $k \in \mathbb{R}$

$$u(x,t) = \int_{\infty}^{\infty} A(k) e^{i(kx - w(k)t)} dk,$$

where $A(k)$ is the Fourier transform of $u(x,0)$. Notice that $\omega = k - k^3$ implies that $\omega_k = 1 - 3k^2$. This relation gives us the group velocity - the velocity of a wave packet. It's also the velocity of the progagation of energy.

$$c_g = 1 - 3k^2.$$

Here, we have $\omega : \mathbb{R} \to \mathbb{R}$, but this is only true if we add a term with odd derivatives. If we instead consider

$$u_t + u_x - u_{xx} = 0,$$

we arrive at the dispersion relation $\omega = k - ik^2$. Then for $k \in \mathbb{R}$, we have

$$u(t,x) = e^{-k^2 t} e^{ik(x-t)}$$

Which describes a wave that decays as $t \to \infty$ for $k \neq 0$, and moves at a speed of unity for all $k \in \mathbb{R}$. This type of wave decay is called dissipation.

The standard wave equation is generally only valid for small amplitudes. A better approximation in many cases is

$$u_t + (1+u)u_x = 0$$

This includes the simplest type of nonlinearity (uu_x). Using the method of characteristics, we have

$$\frac{dx}{dt} = 1 + u \implies \int dx = \int (1+u) dt \implies x = (1+u)t + x_0$$

Along these curves,

$$\frac{du}{dt} = 0 \implies u(t,x) = f(x) \implies u(0,x_0) = f(x_0) = f(x - (1+u)t)$$

$$\implies u(t,x) = f(x - (1+u)t)$$

Such a wave (say, for $f > 0$) is single valued for finite time, but will eventually become multivalued and "break" (like an ocean wave). We can also consider nonlinear wave equations with dispersive or dissipative terms, such as the Korteweg-de Vries equation (nonlinearity + dispersion):

$$u_t + (1+u)u_x + u_{xxx} = 0$$

Or the Burgers equation (nonlinearity + dissipation):

$$u_t + (1+u)u_x - u_{xx} = 0$$

For the KdV equation, we can consider the transformations

$$1+u \to \alpha u, \qquad t \to \beta t, \qquad x \to \gamma x$$

where α, β, γ are nonzero real constants, to yield

$$u_t + \frac{\alpha\beta}{\gamma} u u_x + \frac{\beta}{\gamma^3} u_{xxx} = 0.$$

A common convention is

$$u_t - 6uu_x + u_{xxx} = 0.$$

Note that higher dimension KdV equations exist (ring waves, waves crossing obliquely).